Javad Komijani

Senior Computational Physicist

Zurich, Switzerland
+41 76 796 9992
jkomijani@phys.ethz.ch
jkomijani.github.io

Professional Summary

Senior Computational Physicist with over 10 years of experience in scientific computing and 5 years in scientific machine learning. Develops generative models (diffusion models, normalizing flows) in PyTorch for large-scale simulations. Expert in differential equations, Bayesian inference, and numerical optimization. Bridges advanced mathematical modeling with machine learning to solve complex computational problems.

Technical Skills

Programming	Python, C/C++, Cython, Bash
ML	PyTorch, Generative Models, Diffusion Models, Scientific ML
Applied Math	Differential equations, Bayesian inference, Monte Carlo, Optimization
Systems	Linux, HPC clusters, Parallel computing, Git
Data	NumPy, SciPy, Pandas, Scikit-learn, SQL

Professional Experience

2021–Present **Senior Postdoctoral Researcher**, *ETH Zurich*

- Designed and implemented GPU-based generative models for large-scale lattice simulations.
- Developed custom PyTorch extensions for differentiable linear algebra and ODE solvers.
- Executed computational workflows on HPC clusters.
- Contributed ML modules to the interTwin Digital Twin Engine; supervised MSc/PhD students.

2015–2021 **Postdoctoral Researcher**, *TU Munich / Glasgow / Tehran*

- Applied Bayesian inference and Monte Carlo simulations to large-scale numerical datasets.
- Developed scalable lattice-QCD simulation pipelines in HPC environments.
- Collaborated internationally in computational and mathematical physics projects.

Selected ML & Scientific Computing Projects

Normflow	Normalizing Flows in PyTorch; supporting complex high-dimensional scientific distributions.
Diffusion	Diffusion-based generative models for structured scientific data on Lie group manifolds.
Linear Alg.	Linear Algebra Extension for PyTorch; supporting automatic differentiation for eigenvalue and singular value decompositions of unitary matrices.
Differentiable ODE Solver	Adjoint-based ODE solver extension supporting automatic differentiation for scientific ML workflows.

Education

2015 **PhD in Physics**, *Washington University in St. Louis*

2009 **MSc in Electrical Engineering**, *University of Tehran*

Publications

Selected peer-reviewed publications in computational physics and scientific ML. Full list available upon request.

Languages

English	Fluent
German	Elementary